

ABSTRACT

The growing global demand for electric vehicles (EVs) as a solution to environmental concerns has brought attention to the importance of efficient battery management systems. The performance and longevity of an EV's battery directly impact its driving range, charging time, and overall operational efficiency. Current battery management systems (BMS) are limited to monitoring battery parameters like voltage, temperature, and state of charge (SOC) but lack the capability to predict future battery performance or optimize charging strategies. This project aims to develop an intelligent, machine learning- based battery management system that predicts battery health and optimizes fast charging processes for electric vehicles. The core objective of this system is to leverage machine learning algorithms to analyze real-time battery data, such as voltage, current, temperature, and SOC, and forecast the battery's future health and charging needs. By analyzing historical and real-time data, the system can predict the rate of battery degradation, estimate the remaining battery life, and recommend optimal maintenance schedules. Additionally, it provides real-time insights for fast charging, enabling adaptive charging strategies that balance charging speed with battery longevity.

The system uses supervised learning models, particularly regression analysis and time-series forecasting techniques, to predict future battery performance. These models are trained on large datasets to identify patterns and correlations between various battery parameters. The results allow the system to provide actionable recommendations for users, improving battery efficiency, extending battery life, and preventing unnecessary degradation. The solution offers several advantages over traditional battery management systems, including predictive maintenance, personalized charging recommendations, and an adaptive fast charging mechanism. This approach enhances the overall user experience by maximizing battery lifespan and minimizing the risks associated with fast charging. Ultimately, the integration of machine learning into battery management has the potential to revolutionize the electric vehicle industry by improving performance, reducing maintenance costs, and accelerating the adoption of sustainable transportation.

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CHAPTER 1

INTRODUCTION

In the face of rising global concerns regarding climate change and air pollution, the need for more sustainable modes of transportation has never been greater. Over the last decade, electric vehicles (EVs) have emerged as a promising solution to reduce greenhouse gas emissions, lower dependence on fossil fuels, and improve urban air quality. Unlike traditional internal combustion engine (ICE) vehicles, EVs rely on battery systems to power electric motors, with the most common technology being Lithium-Ion (Li-Ion) batteries. These batteries are crucial to the overall performance of the vehicle, as they determine the driving range, charging time, and overall operational efficiency of the EV. However, while EV adoption continues to rise, there are still significant challenges surrounding the management of battery health and the optimization of charging systems. Battery health and performance are pivotal in determining the efficiency and longevity of electric vehicles. The battery is the most expensive component of an EV, and its degradation directly impacts the vehicle's range, charging time, and ultimately its cost-effectiveness.

Several factors contribute to battery degradation, such as charge cycles, temperature variations, voltage fluctuations, and the State of Charge (SOC), which are continuously changing during the operation of the EV. This dynamic behavior requires a sophisticated battery management system (BMS) that can monitor and regulate these parameters to ensure the battery operates optimally throughout its lifecycle. Inadequate management of these factors can lead to rapid battery degradation, higher maintenance costs, and reduced driving range. One of the challenges faced by conventional BMS is its reactive nature. These systems typically function by detecting predefined thresholds for critical parameters, such as temperature or voltage, and alerting users when an anomaly is detected.

While this approach helps in preventing immediate failure, it doesn't offer insights into the future health of the battery or provide proactive maintenance recommendations. Moreover, as the EV market continues to grow, traditional systems are becoming increasingly inadequate in handling the demands for higher battery efficiency and faster charging times. Another aspect that adds complexity to EV battery management is the need for fast charging. As electric vehicle adoption increases, so does the demand for efficient, high-speed charging solutions. Fast charging provides the convenience of significantly reducing downtime, enabling users to quickly recharge their vehicles during short stops. However, the rapid charging process can exert excessive strain on the battery, potentially leading to

overheating, reduced battery lifespan, and safety hazards.

The challenge here lies in balancing charging speed with battery health, ensuring that the battery is charged quickly without compromising its longevity. Fast charging typically involves applying higher currents than regular charging, which can cause heat generation within the battery. Prolonged exposure to high temperatures can result in the breakdown of the battery's internal chemistry, accelerating capacity loss. Therefore, without a dynamic charging system that takes into account the current state of the battery, external temperature, and charging patterns, the risk of damaging the battery through fast charging remains high. There is also a need for an intelligent system that can predict the optimal charging duration and current to ensure that fast charging does not interfere with the battery's overall health. To address these challenges, there is a growing demand for more intelligent systems that can provide predictive insights into battery performance, health, and charging need.

Traditional battery management systems only focus on monitoring the present state of the battery, without considering its future condition. However, with advancements in data analytics and machine learning (ML), there is an opportunity to create a predictive battery management system that can forecast battery degradation over time and suggest maintenance or replacement schedules based on real-time data. Machine learning algorithms, especially supervised learning and time-series analysis, have the capability to predict future battery behavior by analyzing historical data, which includes charge cycles, temperature fluctuations, and charging times. By using real-time data from the EV's battery, such a system can accurately predict the battery's remaining lifespan, estimate future degradation, and recommend the best charging practices. This predictive capability can significantly reduce the risk of unexpected failures, reduce maintenance costs, and improve the overall user experience.

Moreover, the use of machine learning can also help in optimizing charging strategies, particularly in the context of fast charging. By analyzing parameters such as battery SOC, temperature, and driving conditions, an ML algorithm can recommend the most efficient and safe charging parameters for the battery, thereby minimizing damage while still providing the benefits of faster charging. The application of machine learning to battery management is not just a futuristic concept; it has already begun to revolutionize industries such as consumer electronics and electric vehicles. Machine learning techniques, such as decision trees, neural networks, and regression analysis, can help develop models that understand complex relationships between various battery parameters and provide real-time insights into battery health. These models learn from vast amounts

of historical and real-time data and can evolve over time, becoming more accurate and reliable in predicting battery behavior and recommending actions. In the context of EV battery management, ML models can be trained to recognize patterns that are indicative of battery degradation, such as voltage drops, charging inefficiencies, or temperature spikes.

These models can then provide alerts to the user about potential issues and predict the remaining useful life of the battery, allowing for better maintenance scheduling and reducing the risk of unexpected failures. For fleet operators, these systems can also predict the optimal replacement schedules for batteries, ensuring that the fleet remains operational and efficient at all times. Machine learning also plays a significant role in fast charging prediction. Traditional fast-charging algorithms apply a standard charging rate, regardless of the battery's condition. By integrating machine learning, it is possible to develop more adaptive charging algorithms that adjust the charging current based on the battery's health, temperature, and state of charge. This personalized approach ensures that fast charging is safe and efficient, minimizing the risks associated with high-current charging and prolonging battery life. As the global adoption of electric vehicles accelerates, the need for sophisticated, intelligent battery management systems becomes even more pressing.

Current systems are insufficient to handle the growing demands for high-performance batteries and fast-charging infrastructure. Therefore, the need for a data-driven, machine learning-based solution to manage EV battery health, performance, and charging time is more important than ever. Not only will this innovation provide a better user experience by improving battery longevity and efficiency, but it will also significantly contribute to sustainable transportation by maximizing the performance of EVs and minimizing waste from early battery replacements. In addition, the development of intelligent charging infrastructure is essential to meeting the growing demand for faster and more efficient charging across the globe. This project aims to address these challenges by combining real-time data collection with predictive machine learning algorithms to develop a smart battery management system. This system will help optimize both battery performance and fast charging, providing valuable insights into battery health, extending battery life, and improving the overall efficiency of electric vehicles.

The introduction of machine learning into the field of electric vehicle battery management and fast charging optimization is a natural progression in the ongoing effort to enhance the sustainability and efficiency of electric vehicles. As the demand for EVs continues to rise, addressing challenges related to battery health, charging time, and efficiency is critical. By leveraging machine learning to provide predictive maintenance and optimize charging strategies, this project offers a smart, adaptive solution that will not only extend the lifetime of EV batteries but also improve the overall performance and user experience. The integration of such a system into both individual EVs and commercial fleets holds the potential to revolutionize how we manage battery health and charging in the electric vehicle industry.

1.1 Project Motivation

The rapid adoption of electric vehicles (EVs) has brought significant attention to the performance, efficiency, and longevity of battery systems. However, challenges such as battery degradation, inefficient energy usage, and unpredictable charging behavior remain unresolved. These issues not only affect the reliability of EVs but also impact user experience and the wider acceptance of electric mobility. Therefore, there is a critical need to develop intelligent systems that can manage and monitor battery performance in real-time, ensuring safety, efficiency, and extended battery life.

With the emergence of smart sensors and machine learning, we now have the opportunity to collect meaningful data from EV battery packs—such as voltage, current, temperature, and state-of-charge (SOC)—and use predictive models to gain insights into battery health and charging behavior. Machine learning models can analyze patterns and predict fast charging feasibility, which helps avoid unsafe charging conditions and enhances battery management. Such predictive analysis also enables preventive maintenance and reduces the risk of battery failure.

This project aims to integrate sensor-based data acquisition with a machine learning-driven Battery Management System (BMS), coupled with a user-friendly interface for real-time monitoring and alerts. By implementing this intelligent and affordable solution, the project contributes to making EVs more reliable, sustainable, and accessible, especially in emerging markets where cost-effective innovations are essential. The motivation is not only technological advancement but also environmental impact through smarter energy usage and reduced e-waste.

1.2 Project Contribution

- **Smart Battery Monitoring:** Developed a real-time battery monitoring system using voltage, current, temperature, and SOC sensors to ensure safe and efficient EV operation.
- **ML-Based Charging Prediction:** Implemented a machine learning model to predict fast charging feasibility, reducing battery stress and enhancing charging safety.
- **Data Acquisition Integration:** Designed a low-cost data acquisition unit with Wi-Fi capability to transmit sensor data for cloud-based processing and analysis.
- **User Interface Dashboard:** Created a user-friendly interface that displays battery health insights, alerts, and charging recommendations for improved user awareness.
- **Cost-Effective Prototype:** Delivered a budget-friendly and scalable solution for intelligent battery management, suitable for low-cost EV applications and student research.

1.3 Scope of the project

The project focuses on enhancing the performance, safety, and longevity of electric vehicle (EV) batteries by leveraging real-time monitoring and machine learning-based prediction. As EV adoption rises, the need for smarter battery systems becomes essential. This project addresses that need by proposing a system that not only monitors critical battery parameters but also predicts the feasibility of fast charging, contributing to better energy utilization and vehicle reliability. It enables predictive analysis using machine learning algorithms on collected battery data such as voltage, current, temperature, and state of charge. This analysis helps in identifying patterns and anomalies that may affect battery health, allowing early detection of issues and optimization of charging behavior. By doing so, the project supports preventive maintenance strategies and helps in minimizing battery degradation over time.

The scope also extends to integrating low-cost, IoT-enabled hardware that collects real-time data from the battery and communicates it to a cloud-based analytics platform. This enables remote monitoring and decision-making, making it suitable not only for individual EV owners but also for fleet operators, researchers, and manufacturers looking to gather operational insights from multiple EVs. The proposed system can be further scaled or modified for different types of battery chemistries and EV models, offering flexibility and adaptability. Its modular design allows upgrades and integration with other smart vehicle components, enabling future expansion into full vehicle

energy management systems. This makes the project a foundational framework for more complex BMS innovations. Overall, the project contributes to sustainable EV usage by promoting efficient battery usage, enhancing charging safety, and extending battery life. It aligns with the global move toward cleaner transportation and supports government and industry initiatives targeting carbon neutrality and electric mobility transformation.

CHAPTER 2

PROJECT DESCRIPTION

2.1 Existing System

The existing battery management systems (BMS) in electric vehicles are primarily designed for basic monitoring and protection. They track essential parameters like voltage, current, temperature, and state of charge (SOC) to ensure the battery operates within safe limits. These systems provide real-time feedback and trigger protection mechanisms in case of overvoltage, overcurrent, or overheating. However, they are mostly reactive and lack the intelligence to predict battery degradation or optimize performance over time. Most conventional systems do not employ advanced analytics or predictive models.

Data collected is often used only for immediate control decisions rather than being analyzed to identify long-term usage patterns or predict failures. As a result, battery health is managed conservatively, leading to reduced performance, inefficient charging cycles, and potential safety concerns. These systems also fall short in adapting to diverse driving behaviors and environmental conditions, limiting their effectiveness across varied user scenarios. Additionally, traditional BMS architectures are limited in terms of remote accessibility and scalability. They usually do not integrate with cloud-based platforms or IoT frameworks, which restricts their use for fleet-level analysis or remote diagnostics. With no machine learning capabilities, they cannot recommend optimized charging patterns or forecast fast charging viability, which are essential for improving battery life and user convenience in modern electric vehicles.

2.2 Proposed System

The proposed system introduces a smart battery management approach that integrates real-time sensor data acquisition with machine learning-based predictive analytics. It continuously monitors critical parameters like voltage, current, temperature, and state of charge (SOC) from the battery using sensors connected to a microcontroller. These real-time values are processed and analyzed to detect anomalies, predict battery performance, and assess fast charging feasibility, offering a proactive solution over traditional BMS. Unlike existing systems, this model utilizes AI/ML algorithms trained on historical and real-time data to predict battery health trends, degradation patterns, and optimal charging conditions.

By forecasting the battery's response to various charging methods, the system can suggest the most efficient and safe charging strategy. This not only improves battery efficiency but also prevents thermal stress and early aging, ultimately extending battery lifespan. The system also supports remote monitoring and decision-making through cloud integration, making it scalable for large-scale applications like EV fleets. Its modular design allows customization for different vehicle models and battery chemistries. The combination of IoT, real-time monitoring, and intelligent prediction transforms the BMS into a data-driven platform that enhances safety, performance, and sustainability in electric vehicles.

2.3 Working Principal

The working principle of the proposed project, “Electric Vehicles Battery Management and Fast Charging Prediction Analysis Using Machine Learning,” is based on real-time monitoring, intelligent data processing, and predictive analysis to optimize battery usage and ensure safety, longevity, and efficient charging in electric vehicles (EVs). The system integrates both software and hardware components to collect, analyze, and act upon battery-related parameters such as voltage, current, temperature, and state of charge (SOC). The hardware setup begins with sensors that measure critical battery parameters. Voltage sensors, current sensors (like Hall-effect sensors), and temperature sensors are interfaced with a microcontroller (such as an ESP32 or Arduino).

These sensors constantly collect data from the battery pack and send it to the microcontroller for initial filtering and preprocessing. The microcontroller then transmits this data to a local or cloud-based system for further analysis. The setup may include a wireless transmission module (e.g., Wi-Fi/Bluetooth) to support real-time remote monitoring. Once the data is collected, the software module takes over. This part consists of machine learning models trained on a dataset of EV battery performance metrics. The model learns patterns from historical data, such as how voltage drops over time, how temperature affects performance, and how the battery behaves during fast charging. Using regression or classification models (e.g., Random Forest, Decision Tree, or Support Vector Machines), the system predicts the battery's future state—its health, efficiency, and safety status.

An important component of the system is the fast charging prediction mechanism. Based on current battery conditions and historical behavior, the model forecasts whether the battery can safely undergo fast charging without degradation. If the algorithm identifies that the battery is not in optimal condition, it advises a slower, safer charging cycle. This prevents thermal runaway, overcharging, or excessive wear, which are common risks in conventional charging systems. Alerts

and visualizations are also generated as part of the system's output. If anomalies like abnormal voltage spikes or temperature rises are detected, the system immediately triggers alerts via a dashboard or mobile application. This allows users or fleet managers to take corrective action before damage occurs. These alerts are customizable and can be adjusted according to the type of vehicle, battery capacity, or user preference.

In addition to individual battery analysis, the system can be scaled up for fleet management. By storing and analyzing data from multiple vehicles, fleet operators can identify usage patterns, optimize charging schedules, and plan maintenance ahead of failures. This makes the system highly applicable for industries involved in logistics, public transport, or commercial EV fleets. Overall, the project works by combining real-time hardware monitoring with intelligent software analysis. The synergy of IoT, sensor data, and machine learning enables the system to actively monitor battery performance, predict faults, support fast charging decisions, and improve overall efficiency. This not only enhances the reliability and lifespan of EV batteries but also contributes to safer and smarter electric transportation.

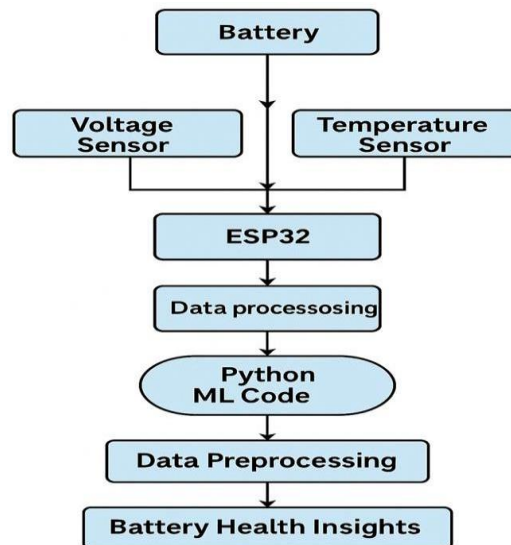


Figure 2.3 Block Diagram

2.4 Integration with cloud

Cloud integration in the context of an Electric Vehicle Battery Management and Fast Charging Prediction System involves connecting the hardware components like sensors and microcontrollers

(e.g., ESP8266) to a cloud platform. This connection allows real-time battery data such as voltage, current, temperature, and State of Charge (SOC) to be uploaded, stored, analyzed, and visualized remotely. The cloud acts as a central repository where all data from various EVs can be accumulated for intelligent processing and decision-making. The ESP8266 Wi-Fi module facilitates wireless communication between the battery management system and the cloud. When programmed with appropriate firmware, the ESP8266 reads sensor data and transmits it via HTTP or MQTT protocols to a cloud server.

This server could be hosted on platforms like Firebase, Blynk, or ThingSpeak, which are simple and beginner- friendly for IoT applications. Real-time data monitoring allows the system to detect abnormalities in battery parameters. If a temperature spike or voltage drop is detected, an alert can be triggered and displayed on a cloud dashboard or sent as a notification to the user's smartphone. This ensures proactive maintenance and enhances battery safety and longevity. The cloud platform can also store historical battery data, which is essential for implementing machine learning models. These models analyze trends over time to predict battery health, estimate remaining useful life, and suggest optimal charging patterns. The integration of AI/ML in the cloud improves the reliability and intelligence of the system.

With cloud connectivity, users, manufacturers, and service centers can access the data remotely, enabling centralized monitoring of fleet batteries. This is particularly useful for logistics companies and electric mobility providers who manage multiple electric vehicles. A web or mobile application linked to the cloud platform provides user-friendly interfaces for visualizing key battery metrics. Dashboards can include graphs showing real-time and historical trends in SOC, voltage, temperature, and current. These insights help users make informed decisions about charging behavior and battery usage. Cloud integration also allows over-the-air updates to firmware and system configurations. This means improvements in ML models or sensor calibration can be pushed remotely without needing physical access to each EV unit. It streamlines maintenance and ensures all units are running the latest version. Security is another essential aspect of cloud integration.

The system must ensure secure communication using protocols like HTTPS and implement authentication methods like API keys or OAuth to restrict data access. This prevents unauthorized users from tampering with sensitive battery data. Scalability is one of the biggest advantages of cloud integration. Whether managing 10 or 10,000 EVs, the cloud infrastructure can be scaled based on demand. This makes the system future-ready as the electric vehicle market continues to grow. Finally,

integrating cloud platforms in this project adds immense value by enabling real-time monitoring, smart data analytics, remote accessibility, and predictive maintenance. It enhances the performance, safety, and lifespan of electric vehicle batteries, supporting sustainable mobility.

CHAPTER 3

HARDWARE DESCRIPTION

3.1 Voltage Sensor

A voltage sensor is a device that detects and measures the electrical potential difference between two points in a circuit. It plays a vital role in monitoring power systems, especially in energy management and battery monitoring applications. In this project, the voltage sensor helps track the voltage levels of the battery in real-time, ensuring the battery operates within safe limits. It typically outputs a scaled-down analog voltage signal that corresponds to the actual battery voltage, which can be read by an analog pin on the NodeMCU. This data is essential for the machine learning model, which uses voltage as a primary input feature to predict battery health and estimate charging time. The sensor used here is connected directly to the battery terminals, and its output is connected to the analog input of the NodeMCU. It does not necessarily require a breadboard unless additional components like resistors or filters are added. Accurate voltage monitoring is crucial to avoid overcharging or undercharging, which can degrade battery life. By constantly supplying voltage readings, this sensor enables smarter and more efficient battery management in electric vehicles.



Figure 3.1 Voltage Sensor

3.2 Current Sensor

A current sensor is used to detect and measure the flow of electric current in a wire or circuit and convert it into a measurable signal. The sensor's general function includes measuring AC or DC current and producing a proportional output that can be interpreted by microcontrollers. In this project, the current sensor monitors the charging and discharging current of the electric vehicle battery. This information is critical for managing the battery efficiently, as it allows the system to

determine the real-time power usage and input this data into the prediction model for fast charging analysis. The sensor typically works using Hall- effect principles, providing electrical isolation and safe interfacing with the NodeMCU. Since the setup avoids a breadboard, the current sensor can be connected directly using jumper wires to simplify wiring. Monitoring current is important to prevent overloading and overheating, which can degrade the battery. The sensor helps the AI model learn how current patterns affect battery performance over time, improving prediction accuracy. Overall, the current sensor plays a vital role in ensuring safe, efficient, and intelligent battery management.

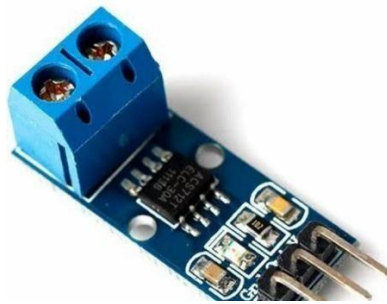


Figure 3.2 Current Sensor

3.3 Temperature Sensor

A temperature sensor measures the ambient or object temperature and provides a corresponding electrical signal. It is a vital component in systems where thermal behavior influences performance or safety. In this project, the temperature sensor monitors the battery temperature during charging and discharging cycles. Monitoring temperature is essential because overheating can significantly reduce battery life and increase safety risks. The sensor works by producing a voltage output that varies with temperature, which is read by the NodeMCU's analog input. Common temperature sensors like the LM35 or DHT11 are used for their ease of use and compatibility with microcontrollers. The sensor data helps the system decide whether the battery is within a safe operating range. This temperature data is also used in the machine learning model to correlate thermal behavior with battery efficiency and charging needs. Even without a breadboard, the sensor can be securely connected with jumper wires for a compact and reliable setup. Accurate temperature readings ensure thermal protection and help optimize the charging algorithm to extend battery life. The sensor thus acts as a protective and analytical tool, enhancing the overall intelligence of the battery management system.



Figure 3.3 Temperature sensor

3.4 NodeMCU

NodeMCU is a low-cost, open-source microcontroller development board based on the ESP8266 Wi-Fi module. It is widely used in IoT projects due to its built-in Wi-Fi connectivity and compatibility with the Arduino IDE. In this project, the NodeMCU acts as the brain of the system, gathering data from sensors (voltage, current, temperature) and displaying it on the OLED screen. It also enables data transfer to cloud platforms or AI-based systems for predictive analysis. The NodeMCU reads analog and digital inputs from connected sensors and processes this data in real-time. It then sends relevant data to an external AI model (or cloud platform) and receives output such as charging suggestions or battery health insights. It also handles communication with the OLED display to show live sensor readings. The board's compact design makes it ideal for direct wiring without a breadboard. With just USB power and a few jumper wires, it can be connected and programmed quickly. The NodeMCU's ease of integration, programmable control, and networking capabilities make it central to the success of this battery management system, serving as the bridge between physical sensors and intelligent decision-making.



Figure 3.4 NodeMCU

3.5 OLED Display

An OLED (Organic Light Emitting Diode) display is a flat-panel screen that uses organic compounds to emit light when electric current passes through. It is lightweight, power-efficient, and offers high contrast, making it ideal for embedded systems. In this project, the OLED display is used to visually present real-time data such as voltage, current, temperature, and battery status. It provides a user-friendly interface to monitor battery health directly from the hardware without needing a PC or smartphone. The display typically communicates with the NodeMCU over the I2C protocol, using just two wires (SCL and SDA), which makes it easy to connect even without a breadboard. The NodeMCU updates the display continuously with the latest readings from the sensors. This visual feedback is essential during testing and deployment to ensure the system is functioning correctly. The display helps users quickly assess the battery's condition, detect faults, and understand prediction outcomes. OLED's small size and low power consumption make it perfect for portable systems like EV battery monitors. In summary, the OLED serves as a compact and efficient way to communicate real-time data directly to the user.



Figure 3.5 OLED Display

3.6 Power Supply

The power supply provides the necessary electrical energy for the operation of all the electronic components in the system. In this project, a regulated 5V or 3.3V power source is used to power the NodeMCU and connected sensors. A USB connection, lithium-ion battery, or an adapter-based power module can be used. Consistent and safe power delivery is critical, especially in a compact system where components are directly connected without a breadboard. The power supply ensures that sensors provide accurate readings and that the NodeMCU operates without reset or malfunction. In many setups, a portable power bank is used for mobility. For bench testing, USB from a laptop or an adapter is convenient. The power supply may also be derived from the EV battery in future real-time applications, but must be regulated using a buck converter. Clean and sufficient power ensures the sensors work without noise or instability, and that Wi-Fi communication is reliable. If the power fluctuates, it may lead to incorrect data collection, affecting AI predictions. Hence, in this system, a stable power supply ensures smooth and continuous monitoring and analysis.



Figure 3.6 Power Supply

3.7 Battery

The battery is the central component of any electric vehicle and energy storage system. In this project, the battery under observation serves as the data source for voltage, current, and temperature analysis. Its performance and health are monitored using sensors connected to the NodeMCU. Although a real EV battery is complex and large, this setup uses a smaller-scale lithium-ion or lead-acid battery to replicate the working of a typical EV battery system. The sensors track charging behavior and help detect degradation, overheating, or unusual discharges. The battery's voltage is sensed to calculate the state of charge (SoC), while the current sensor monitors energy flow during charging or usage. The temperature sensor ensures it doesn't overheat. This data is then used by a machine learning model to predict when fast charging is safe and how long the battery will last under different conditions. While the current hardware does not use a breadboard, the battery is safely connected through a voltage sensor to prevent over-voltage damage to the NodeMCU. The battery acts both as the object of analysis and the energy source, making it the focal point of the entire system.



Figure 3.7 Lithium Battery

CHAPTER 4

SOFTWARE DESCRIPTION

4.1 **Arduino**

Arduino IDE is an open-source, cross-platform integrated development environment used for writing, compiling, and uploading code to Arduino-compatible boards. Designed to work primarily with microcontroller-based hardware like the Arduino Uno, Mega, and NodeMCU (ESP8266/ESP32), the IDE is based on C and C++ languages, making it suitable for both beginners and experienced developers. The software includes features such as syntax highlighting, automatic indentation, brace matching, and error display to ease programming and debugging. In this project, the Arduino IDE is used to program and manage the ESP8266 NodeMCU microcontroller, which acts as the control unit of the battery management system. Using this IDE, the embedded logic is written to interface with sensors such as voltage, current, and temperature sensors. These sensors collect real-time data from the electric vehicle battery. The IDE facilitates the integration of external sensor libraries and ensures accurate sensor calibration. With Arduino's inbuilt serial monitor and plotting tools, the data can also be visualized during development.

Sensor Data Acquisition and Control

Arduino IDE facilitates seamless integration of analog and digital sensor data through the NodeMCU. For instance, the voltage sensor provides analog voltage levels, which are read using the `analogRead()` function and scaled using basic arithmetic operations. Similarly, the current and temperature values are fetched using digital or analog channels, processed, and stored in variables that are used for decision-making logic in the loop.

Threshold Detection and Display Output

The IDE allows us to program decision-making logic where thresholds for temperature, voltage, and current are defined. If any parameter exceeds safe limits, visual alerts can be triggered via the OLED display. This logic runs continuously in the `loop()` function, ensuring real-time monitoring of battery health. The `setup()` function is used to initialize pin modes, serial communication, and OLED configuration at the beginning of the operation.

Serial Communication and Machine Learning Integration

The Arduino IDE enables serial communication between the NodeMCU and Python-based

machine learning modules on a PC. This is crucial for transmitting real-time sensor data to Python, where a regression model processes it and predicts fast-charging feasibility. The communication occurs over USB or WiFi using serial or HTTP/MQTT protocols. Proper baud rates and communication protocols are defined in the Arduino sketch using `Serial.begin()` and `Serial.print()` functions.

Integration with IoT Platforms

While cloud integration like Blynk or Firebase is not implemented in the first stage, the Arduino IDE can support libraries for WiFi and IoT communication (e.g., `ESP8266WiFi.h`, `BlynkSimpleEsp8266.h`). These libraries make it easy to expand the project into an IoT-enabled system for remote monitoring and control in future iterations.

Library Usage and Compilation

Libraries such as `Adafruit_GFX`, `Adafruit_SSD1306`, and `ESP8266WiFi` are included in the Arduino sketch to handle OLED display operations and potential WiFi functionalities. Once the program is compiled successfully using the “Verify” button, the “Upload” button sends the binary code to the ESP8266 microcontroller. Errors, if any, are displayed at the bottom, making debugging simple.

In conclusion, Arduino IDE acts as the backbone for embedding logic, acquiring sensor data, interacting with machine learning models, and controlling output in this battery management project. Its simplicity, compatibility with numerous libraries, and support for NodeMCU make it the ideal development environment for beginners aiming to build real-world embedded systems.

4.2 Arduino build process Overview:

The Arduino build process begins with preprocessing, where the Arduino IDE transforms the user-written sketch into valid C/C++ code. This includes adding necessary function prototypes and linking user code with the core Arduino functions. The modified code is then passed to the `avr-gcc` compiler, which converts it into object files containing machine-level instructions. These object files are linked with the standard Arduino core libraries that provide predefined functionalities like digital I/O, serial communication, and timing operations. After successful compilation and linking, an Intel Hex file is generated containing the binary code for the microcontroller. This file is uploaded to the Arduino board via a USB or serial interface using

the avrdude tool, which communicates with the bootloader on the microcontroller to store the code in flash memory. Once uploaded, the microcontroller executes the code, beginning with the `setup()` function followed by the repeated execution of the `loop()` function, enabling continuous interaction with sensors and actuators.

4.3 Multi-File Sketches

A sketch in Arduino can be organized into multiple files or tabs to improve readability and manageability, especially for larger projects. These tabs can have extensions such as none (default), `.c`, `.cpp`, or `.h`. Tabs with no extension are concatenated into a single main sketch file before compilation. Tabs with `.c` or `.cpp` extensions are compiled independently as separate source files. Header files with `.h` extensions must be included using `#include "filename.h"` in the sketch to be used. During the build process, the Arduino Environment first inserts `#include "Arduino.h"` at the top of the main sketch file to include core definitions and functions.

It then scans for function definitions and automatically generates corresponding prototypes, which are inserted below any preprocessor directives but above other code. This allows for function use before their actual definitions in the file. However, this auto-generation is limited: it does not handle functions with default parameter values or those within classes or namespaces. To use custom data types or advanced structures, separate header files should be created and included manually. This multi-file organization enables modular design, better code reuse, and collaboration in more complex Arduino projects.

4.4 Blynk Iot Platform

Blynk IoT platform is a powerful and versatile application specifically developed for Internet of Things (IoT) applications. It acts as a bridge between the hardware and the user, allowing real-time monitoring and control of connected devices through a smartphone. Blynk is compatible with various microcontrollers and communication methods, including the widely used ESP8266 module. In this project, Blynk plays a crucial role in facilitating communication between the NodeMCU (ESP8266) and the user via the Blynk mobile application. It enables the display of live data from voltage sensors, current sensors, and temperature sensors connected to the battery system, allowing users to remotely monitor the EV battery status from any location. Blynk also allows data logging and visualization through various widgets such as gauges, charts, and displays, making it easier to understand system performance.

The Blynk ecosystem consists of three main components: the Blynk App, the Blynk Server, and the Blynk Library. The Blynk App, available for Android and iOS, provides a graphical interface where users can create dashboards and add widgets to visualize sensor data and control actuators. The Blynk Server handles the communication between the mobile app and the hardware; it can either be hosted in the cloud or locally. The Blynk Library is installed in the Arduino IDE and helps the ESP8266 board interact with the Blynk server, sending and receiving commands. In this project, the Blynk Cloud is used, which simplifies connectivity and ensures seamless data transmission. Every time a sensor reads data, the ESP8266 sends it to the Blynk Cloud using the library, and from there it gets displayed on the app in real-time. This continuous cycle of sensing, transmission, and visualization enables remote supervision of battery parameters like voltage, current, and temperature, enhancing safety and efficiency in the electric vehicle battery management system.

CHAPTER 5

RESULTS AND DISCUSSION

The implementation of an IoT-based Battery Management System (BMS) using ESP8266 and Blynk platform has produced promising results in enhancing electric vehicle battery monitoring and management. Real-time data collection and visualization through the Blynk app have enabled users to remotely monitor crucial battery parameters such as voltage, current, and temperature. This capability has improved system transparency and user awareness, allowing early detection of abnormalities like overcharging, deep discharging, or thermal issues. The continuous monitoring ensures safe operation, potentially extending the battery's lifespan and preventing damage due to unfavorable conditions.

The system's use of voltage sensors, current sensors, and temperature sensors provided accurate readings that were instantly transmitted via the ESP8266 module to the Blynk cloud. Users could view these readings on their smartphones, facilitating informed decision-making. One of the key findings was the ease of integration and reliability of the ESP8266 in maintaining stable Wi-Fi connectivity, which was essential for uninterrupted data flow. Additionally, the compact nature and low power consumption of the NodeMCU (ESP8266) made it ideal for embedded battery management systems, where space and energy are limited.

The system also demonstrated scalability and adaptability. Since Blynk supports a variety of widgets, users could customize dashboards to suit different use cases. Data trends could be observed over time using graph widgets, which helped in understanding the battery's performance under different load conditions. However, the discussion also brought out some limitations. The accuracy of sensor data could be influenced by environmental factors such as temperature variations or electromagnetic interference. Also, while the system offers real-time monitoring, predictive analytics were not part of the current implementation and would require additional data processing and machine learning integration.

Security remains an area of concern, especially when transmitting sensitive data over the internet. Implementation of authentication protocols and data encryption methods would be necessary for deployment in critical or large-scale applications. Despite these challenges, the ESP8266 and Blynk-based battery monitoring system proved to be an efficient, cost-effective, and user-friendly solution. The results validate its potential in real-world ev applications.

5.1 Hardware Implementation

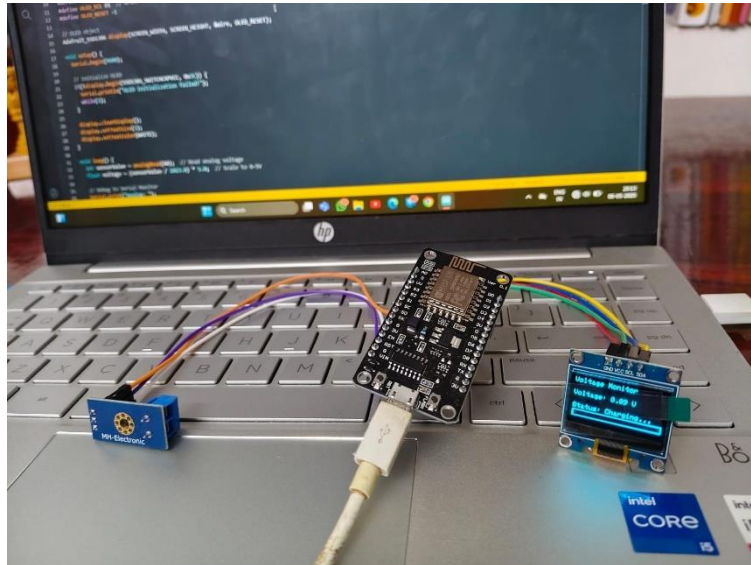


Figure 5.1.1: Hardware Setup

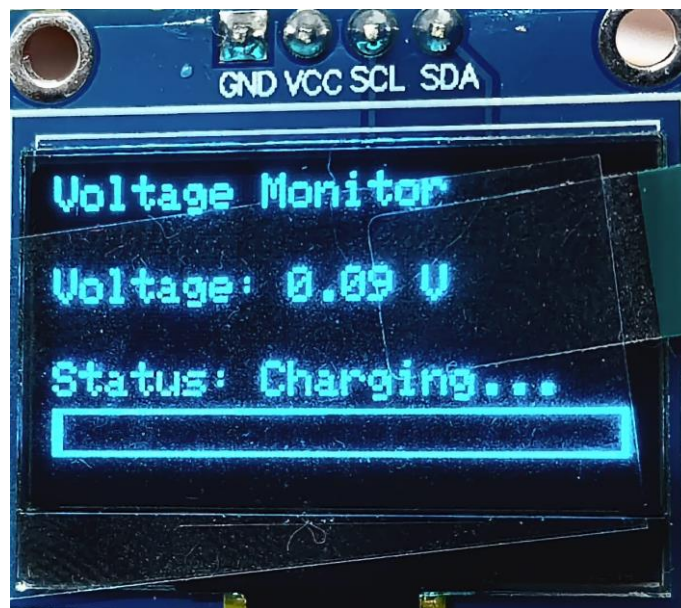


Figure 5.1.2: EV Battery Output

5.2 Simulation and Analysis

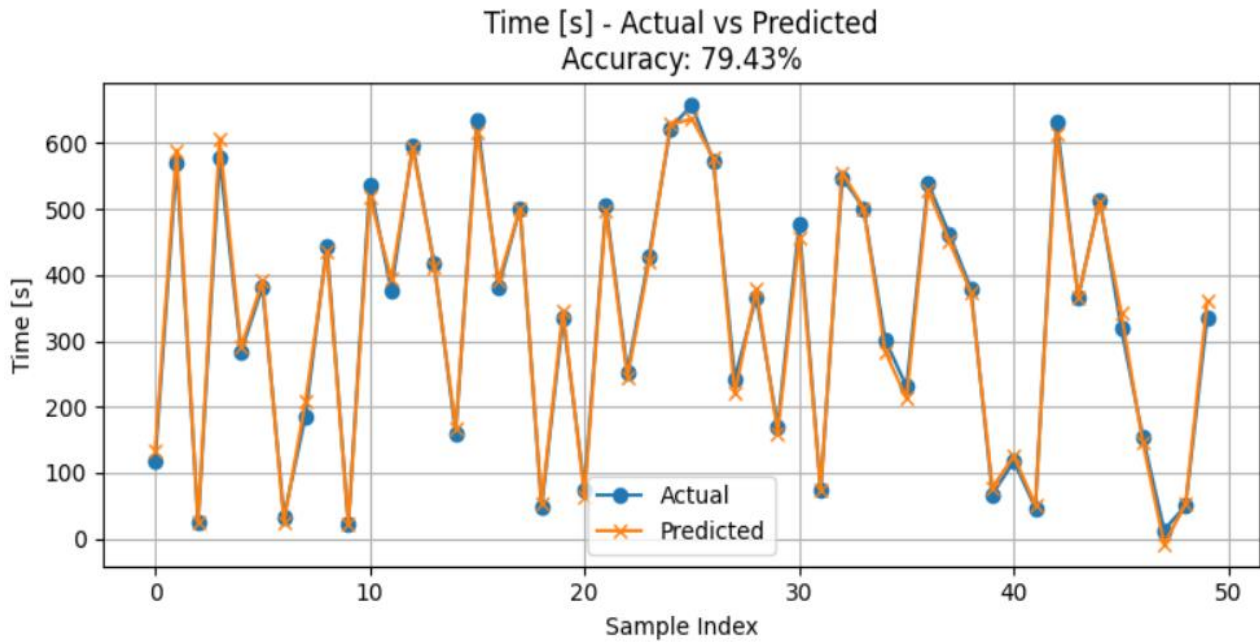


Figure 5.2.1: TIME – ACTUAL VS PREDICTED

S.No	Actual	Predicted
0	118.8	133.562076
1	571.1	588.932270
2	24.7	25.086489
3	577.6	606.794723
4	283.6	290.362324
5	382.1	390.735239

- Mean Squared Error (MSE) for Time [s]: **186.5834**
- Mean Absolute Error (MAE) for Time [s]: **11.3046**
- Root Mean Squared Error (RMSE) for Time [s]: **13.6596**
- Mean Absolute Percentage Error (MAPE): **20.57%**
- Accuracy: **79.43%**
- Simulation Time (Model Training + Prediction): **0.0645 seconds**
- Total Execution Time (Data + Plot + Metrics): **2.6341 seconds**

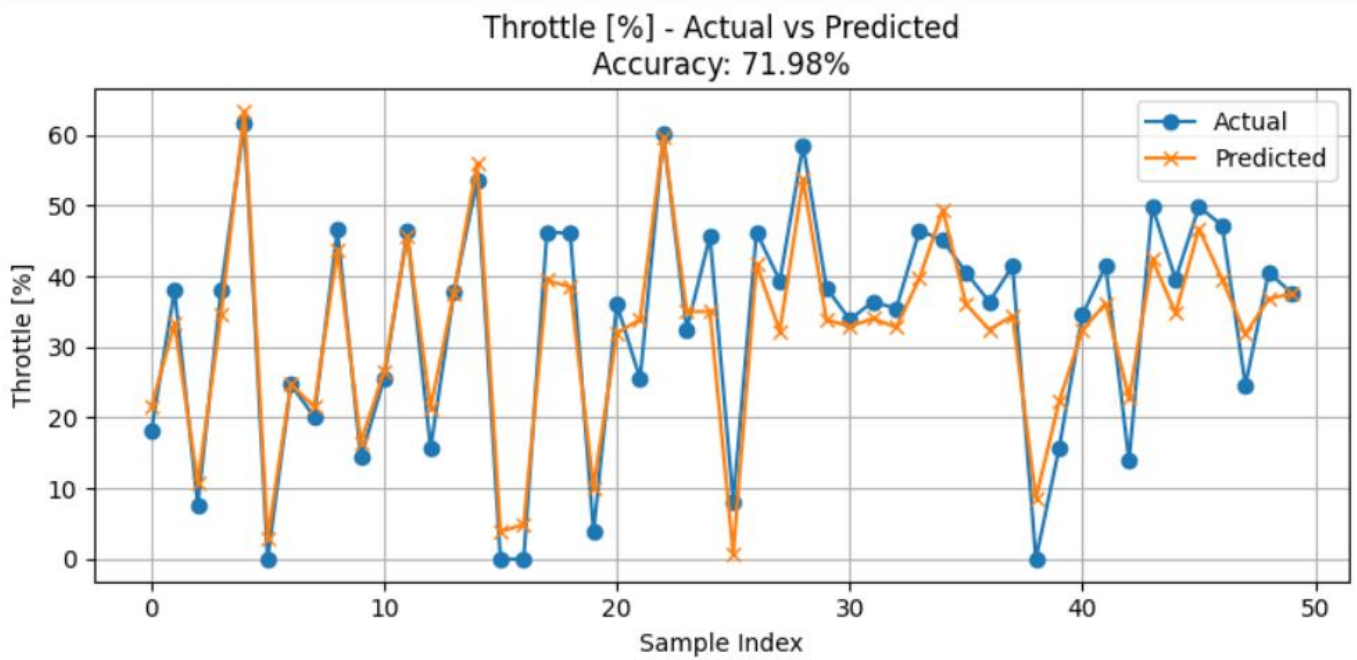


Figure 5.2.2: THROTTLE – ACTUAL VS PREDICTED

S.No	Actual	Predicted
0	18.05	21.470714
1	38.02	33.289372
2	7.60	10.836566
3	38.08	34.550051
4	61.73	63.364225
5	0.00	2.901533

- Mean Squared Error (MSE) for Throttle [%]: **49.0754**
- Mean Absolute Error (MAE) for Throttle [%]: **5.2991**
- Root Mean Squared Error (RMSE) for Throttle [%]: **7.0054**
- Mean Absolute Percentage Error (MAPE): **28.02%**
- Accuracy: **71.98%**
- Simulation Time (Model Training + Prediction): **0.0421 seconds**
- Total Execution Time (Data + Plot + Metrics): **1.1508 seconds**

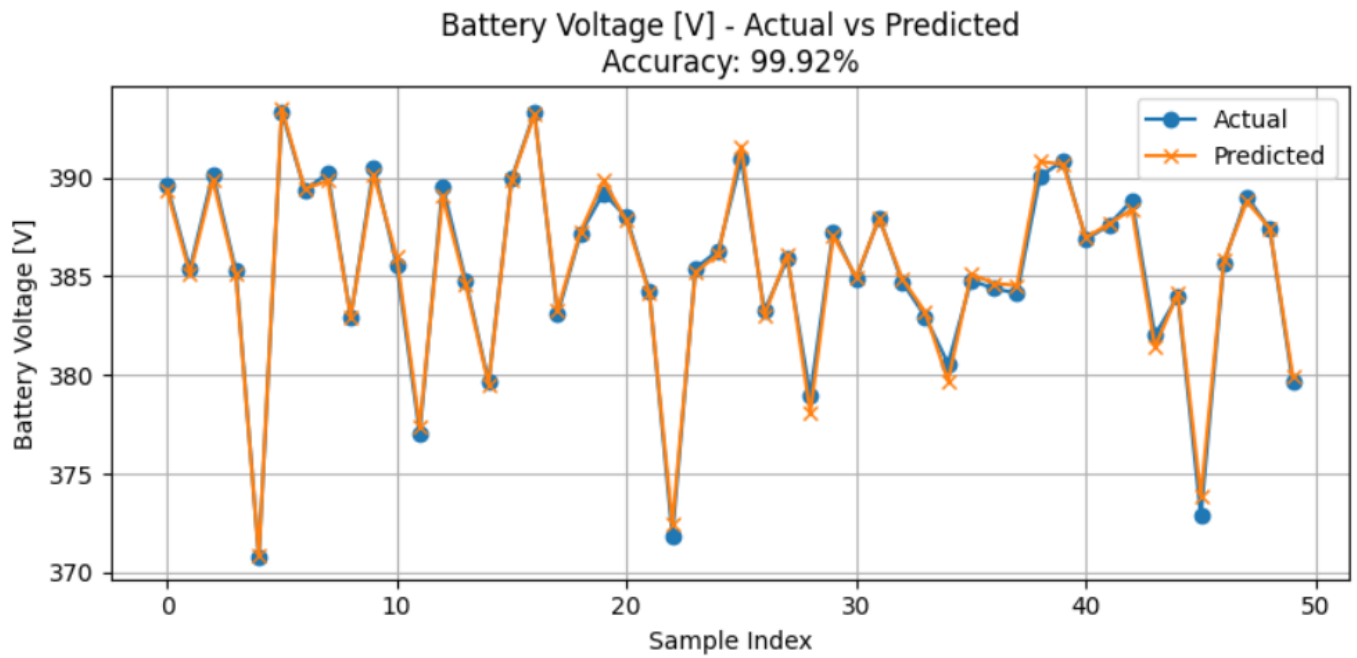


Figure 5.2.3: BATTERY VOLTAGE - ACTUAL VS PREDICTED

S.No	Actual	Predicted
0	389.59	389.374387
1	385.37	385.130136
2	390.16	389.854400
3	385.28	385.084082
4	370.80	370.853490
5	393.30	393.473588

- Mean Squared Error (MSE) for Battery Voltage [V]: **0.1507**
- Mean Absolute Error (MAE) for Battery Voltage [V]: **0.2957**
- Root Mean Squared Error (RMSE) for Battery Voltage [V]: **0.3882**
- Mean Absolute Percentage Error (MAPE): **0.08%**
- Accuracy: **99.92%**
- Simulation Time (Model Training + Prediction): **0.0145 seconds**
- Total Execution Time (Data + Plot + Metrics): **0.3630 seconds**

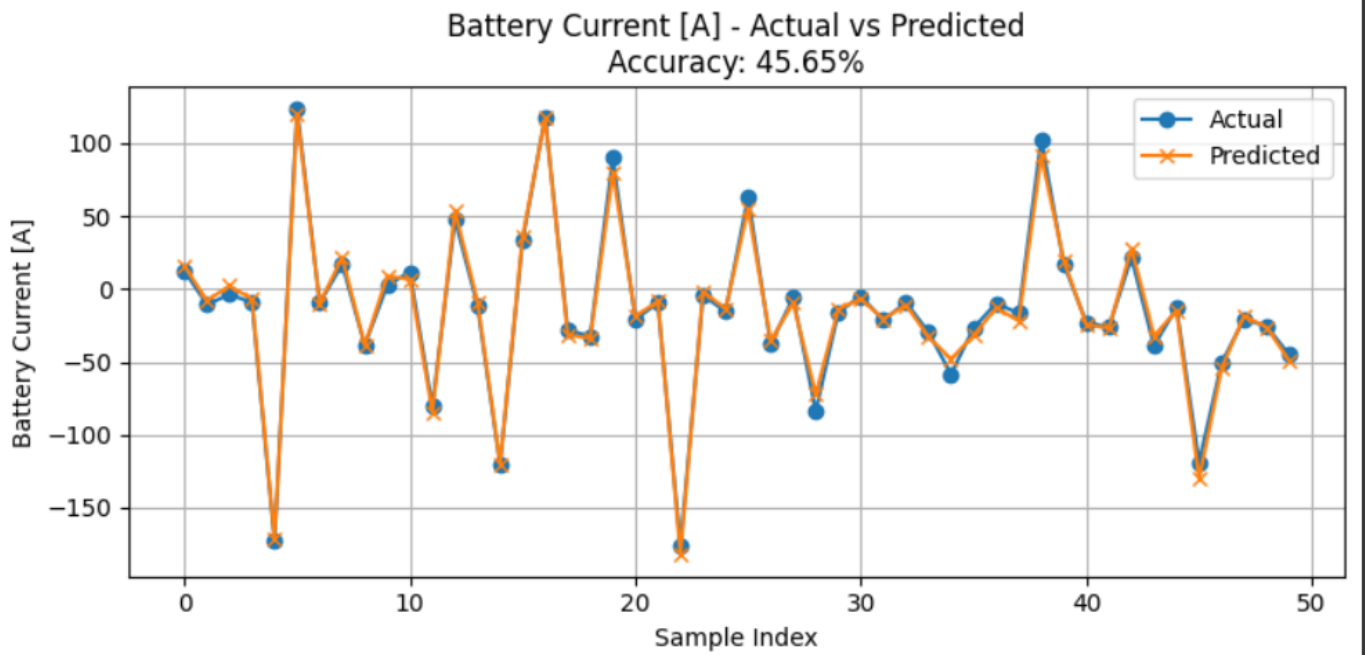


Figure 5.2.4: BATTERY CURRENT -ACTUAL VS PREDICTED

S.No	Actual	Predicted
0	12.42	15.709710
1	-10.69	-7.978010
2	-3.72	2.159821
3	-9.35	-7.044130
4	-172.90	-171.897172
5	123.52	119.494910

- Mean Squared Error (MSE) for Battery Current [A]: **24.8006**
- Mean Absolute Error (MAE) for Battery Current [A]: **3.7663**
- Root Mean Squared Error (RMSE) for Battery Current [A]: **4.9800**
- Mean Absolute Percentage Error (MAPE): **54.35%**
- Accuracy: **45.65%**
- Simulation Time (Model Training + Prediction): **0.0117 seconds**
- Total Execution Time (Data + Plot + Metrics): **0.3066 seconds**

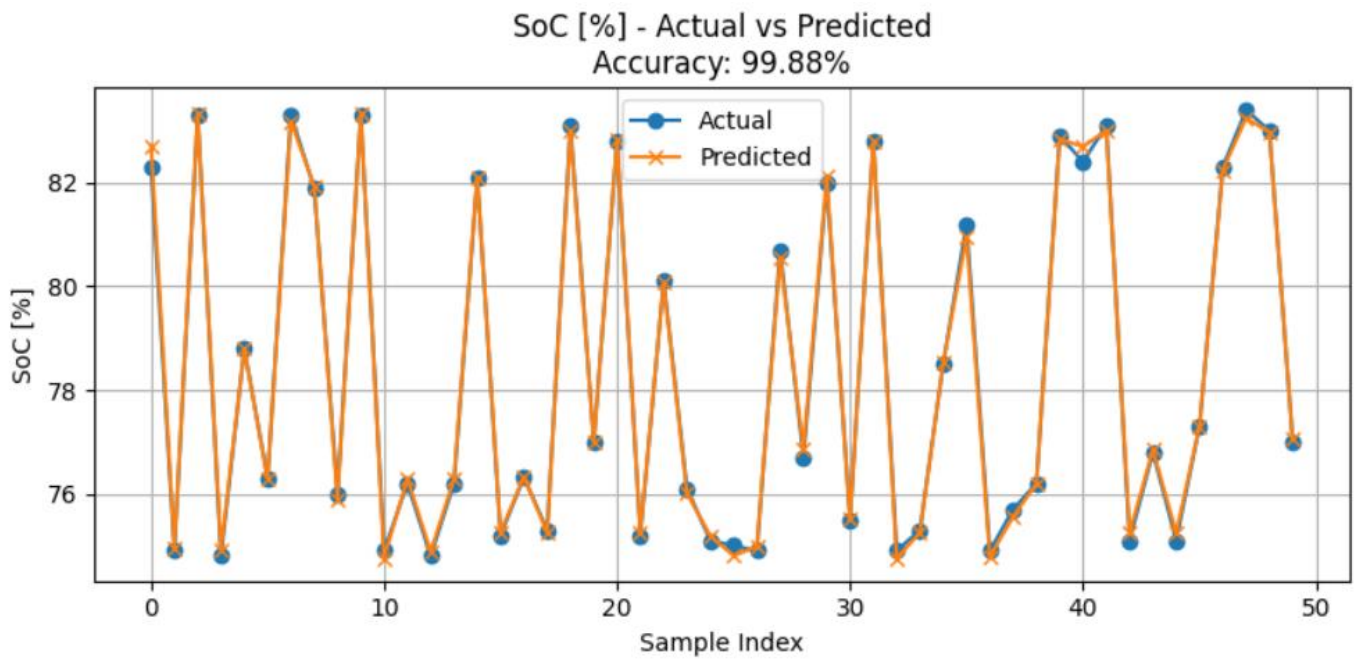


Figure 5.2.5: SOC – ACTUAL VS PREDICTED

S.No	Actual	Predicted
0	82.3	82.712407
1	74.9	74.963921
2	83.3	83.321321
3	74.8	74.916021
4	78.8	78.797783
5	76.3	76.277224

- Mean Squared Error (MSE) for SoC [%]: **0.0133**
- Mean Absolute Error (MAE) for SoC [%]: **0.0935**
- Root Mean Squared Error (RMSE) for SoC [%]: **0.1154**
- Mean Absolute Percentage Error (MAPE): **0.12%**
- Accuracy: **99.88%**
- Simulation Time (Model Training + Prediction): **0.0119 seconds**
- Total Execution Time (Data + Plot + Metrics): **0.2670 seconds**

CHAPTER 6

6.1 Conclusion

The integration of the ESP8266 module into the battery management system for electric vehicles has demonstrated effective performance in real-time monitoring and remote data visualization. Using the Blynk IoT platform, the system efficiently transmitted voltage, current, and temperature readings to a mobile interface, enabling users to monitor battery status from any location. The ESP8266's low power consumption and compact design made it suitable for continuous operation in embedded applications, aligning with the energy efficiency goals of electric vehicles. The module provided reliable Wi-Fi connectivity, ensuring consistent data updates and responsiveness in the Blynk app dashboard.

The cost-effectiveness of ESP8266 contributed to the development of an affordable monitoring solution, allowing scalable implementation in budget-conscious scenarios such as student projects or low-cost electric vehicle systems. The use of sensors for voltage, current, and temperature measurement ensured accurate data acquisition, allowing users to identify potential risks such as overheating, overcharging, or excessive current draw. These capabilities support preventive maintenance and prolong battery life.

Despite its advantages, the system faces challenges such as the need for stable internet connectivity, potential inaccuracies due to sensor noise or environmental fluctuations, and security risks associated with wireless data transmission. Addressing these issues through sensor calibration, data encryption, and reliable power supply design is essential. Overall, the ESP8266-based battery monitoring system presents a practical and efficient approach for improving battery safety, performance, and remote accessibility in electric vehicle applications.

6.2 Future Enhancement

- Integrate support for cloud-based analytics platforms to enable long-term data storage, advanced visualization, and historical trend analysis for battery health monitoring.
- Incorporate over-the-air (OTA) update functionality for the ESP8266 to allow remote firmware updates and system improvements without physical intervention.
- Add mobile app notifications for real-time alerts in case of abnormal voltage, temperature, or current levels to enhance user responsiveness and battery safety.

- Use high-precision sensors and advanced calibration techniques to improve the accuracy of voltage, current, and temperature readings under dynamic load conditions.
- Develop an energy harvesting system (e.g., solar-powered supply) to maintain continuous monitoring capability in off-grid or mobile electric vehicle applications.

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